

**SAFETY DATA SHEET****1.CHEMICAL PRODUCT AND COMPANY IDENTIFICATION**

|                          |                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Product Name</b>      | <b>PENGUIN CEMENT SH-301</b>                                                                                         |
| <b>Manufacturer Name</b> | Sunstar Chemical (Thailand) Co.,Ltd.                                                                                 |
| <b>Address</b>           | 624 Moo 4 Bangpoo Industrial Estate Soi 7A Sukhumvit Rd., Praksa,<br>Muang Samutprakarn, Samutprakarn 10280 Thailand |
| <b>Telephone Number</b>  | +66 2709 4531                                                                                                        |
| <b>Fax Number</b>        | +66 2709 4541                                                                                                        |

**2.HAZARDS IDENTIFICATION****PHYSICAL HAZARDS**

Not classified

**HEALTH HAZARDS**

|                                                    |                                 |
|----------------------------------------------------|---------------------------------|
| Acute toxicity (Oral)                              | Not classified                  |
| Acute toxicity (Dermal)                            | Category 4                      |
| Acute toxicity (Inhalation: Gases)                 | Not applicable                  |
| Acute toxicity: (Inhalation: Vapours)              | Classification not possible     |
| Acute toxicity (Inhalation: Dusts and mists)       | Classification not possible     |
| Skin corrosion/irritation                          | Category 2                      |
| Serious eye damage/eye irritation                  | Category 2A                     |
| Respiratory sensitization                          | Classification not possible     |
| Skin sensitization                                 | Category 1                      |
| Germ cell mutagenicity                             | Category 2                      |
| Carcinogenicity                                    | Category 1                      |
| Reproductive toxicity                              | Not classified                  |
| Specific target organ toxicity - Single exposure   | Not classified                  |
| Specific target organ toxicity - Repeated exposure | Category 1 (Respiratory system) |
| Aspiration hazard                                  | Classification not possible     |

**ENVIRONMENTAL HAZARDS**

|                                                  |                             |
|--------------------------------------------------|-----------------------------|
| Hazardous to the aquatic environment (Acute)     | Category 1                  |
| Hazardous to the aquatic environment (Long-term) | Category 1                  |
| Hazardous to the ozone layer                     | Classification not possible |

**ALLOCATION OF LABEL ELEMENTS****Hazard symbol****Signal word**

Danger

**Hazard statement**

H312 Harmful in contact with skin.  
H315 Causes skin irritation.  
H317 May cause an allergic skin reaction.  
H319 Causes serious eye irritation.  
H341 Suspected of causing genetic defects.  
H350 May cause cancer.  
H372 Causes damage to organs through prolonged or repeated exposure.  
H400 Very toxic to aquatic life.  
H410 Very toxic to aquatic life with long lasting effects.

**Precautionary statements****Prevention**

P201 Obtain special instructions before use.  
P202 Do not handle until all safety precautions have been read and understood.  
P260 Do not breathe dust/fume/gas/mist/vapours/spray.  
P261 Avoid breathing dust/fume/gas/mist/vapours/spray.  
P264 Wash thoroughly after handling.  
P270 Do not eat, drink or smoke when using this product.  
P272 Contaminated work clothing should not be allowed out of the workplace.  
P273 Avoid release to the environment.  
P280 Wear protective gloves/protective clothing/eye protection/face protection.

## Response

P312 Call a POISON CENTER or doctor/physician if you feel unwell.  
P314 Get medical advice/attention if you feel unwell.  
P321 Specific treatment (see label).  
P391 Collect spillage.  
P302+P352 IF ON SKIN: Wash with plenty of soap and water.  
P308+P313 IF exposed or concerned: Get medical attention/advice.  
P332+P313 If skin irritation occurs: Get medical advice/attention.  
P333+P313 If skin irritation or rash occurs: Get medical advice/attention.  
P337+P313 If eye irritation persists: Get medical advice/attention.  
P362+P364 Take off contaminated clothing and wash it before reuse.  
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

## Storage

P405 Store locked up.

## Disposal

P501 Dispose of contents/container to..... (in accordance with local/regional/national/international regulation).

### 3.COMPOSITION/INFORMATION ON INGREDIENTS

**Substance/Preparation** Preparation  
**General Name** Epoxy type Adhesives

**Components Contributing to the Hazard**

| Components    | Cas No     | %WT   |
|---------------|------------|-------|
| Epoxy resin   | 25068-38-6 | 30~40 |
| Calcium oxide | 1305-78-8  | <5    |
| Carbon black  | 1333-86-4  | <10   |

### 4.FIRST AND MEASURES

**Inhalation**

Remove to fresh air and keep at rest in a position comfortable for breathing.

**Skin Contact**

Wash with plenty of soap and water.  
If skin irritation or rash occurs: Get medical advice/attention.

**Eye Contact**

Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.  
If eye irritation persists: Get medical advice/attention.

**Ingestion**

Get medical attention/advice.

### 5.FIRE-FIGHTING

**Extinguishing Media**

Carbon dioxide, dry chemical, water, foam, dry sand

**Special Hazards**

Toxic gases (nitrogen oxides) will foam upon combustion.

**Special Methods**

Evacuate non essential personnel.  
Move container from fire areas if it can be done without risk.  
Keep personnel removed from and upwind of fire.

**Protection of Firefighters**

Wear self-contained breathing apparatus.

### 6.ACCIDENTAL RELEASE MEASURES

**Personal Precautions**

Wear protective equipment and avoid contact with skin and eye.

**Environmental Precautions**

Avoid release to the environment.

**Methods for Cleaning up**

Absorb with liquid-binding materials (sand, sawdust etc.).  
Scoop material into containers for disposal.

## 7.HANDLING AND STORAGE

### Handling

#### Technical Measures

- Do not handle until all safety precautions have been read and understood.
- Do not breathe dust/fume/gas/mist/vapours/spray.
- Wash thoroughly after handling.
- Do not eat, drink or smoke when using this product.
- Contaminated work clothing should not be allowed out of the workplace.
- Keep containers closed when not in use.

#### Precautions

- Use in the well-ventilated areas.

### Storage

#### Suitable Storage Conditions

- Keep container tightly closed.
- Store in a cool (<10°C) and well-ventilated place.
- Protect material from direct sunlight.

## 8.EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Components    | Cas No     | ACGIH (2018,TWA)    |
|---------------|------------|---------------------|
| Epoxy resin   | 25068-38-6 | Not established     |
| Calcium oxide | 1305-78-8  | 2 mg/m <sup>3</sup> |
| Carbon black  | 1333-86-4  | 3 mg/m <sup>3</sup> |

### Personal Protective Equipment

#### Hand Protection

- Protective gloves

#### Eye Protection

- Safety glasses

#### Skin and Body Protection

- Avoid contact with skin.

### Hygiene Measures

- Wash hands before taking a break and at the end of work.

## 9.PHYSICAL AND CHEMICAL PROPERTIES

### Physical State

|             |                 |
|-------------|-----------------|
| Form        | Paste           |
| Color       | Black           |
| Odor        | Slightly odor   |
| pH          | Not established |
| Flash point | >200°C          |
| Density     | About 1.4       |
| Solubility  | Not established |
| Other data  | Not established |

## 10.STABILITY AND REACTIVITY

### Stability

- Stable under the condition of normal usage.

### Possible Hazardous Reactions

- Reactive with amine compounds

### Conditions to Avoid

- Avoid amine compounds (amine, amide etc.)

### Hazardous Decomposition Products

- Nitrogen oxides

## 11.TOXICOLOGICAL INFORMATION

### Acute toxicity

|                 |               |     |                                   |                                                                                                                                         |
|-----------------|---------------|-----|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Epoxy resin :   | LD50 (Oral) : | Rat | 11,400 mg/kg                      | (CERI Hazard Data 2001-36 (2002))<br>(Risk Assessment Report (Food Additives) (F<br>Safety Commission of Japan, 2013))<br>(SIDS (2007)) |
| Calcium oxide : | LD50 (Oral) : | Rat | 5,000 mg/kg and<br>5,916 mg/kg    |                                                                                                                                         |
| Carbon black :  | LD50 (Oral) : | Rat | >8,000 mg/kg and<br>>10,000 mg/kg |                                                                                                                                         |
|                 |               |     |                                   |                                                                                                                                         |

**Skin corrosion/irritation**

|                 |                |
|-----------------|----------------|
| Epoxy resin :   | Category 2     |
| Calcium oxide : | Category 2     |
| Carbon black :  | Not classified |

**Serious eye damage/eye irritation**

|                 |                |
|-----------------|----------------|
| Epoxy resin :   | Category 2B    |
| Calcium oxide : | Category 1     |
| Carbon black :  | Not classified |

**Skin sensitization**

|               |            |
|---------------|------------|
| Epoxy resin : | Category 1 |
|---------------|------------|

**Germ cell mutagenicity**

|               |                |
|---------------|----------------|
| Epoxy resin : | Not classified |
|---------------|----------------|

**Carcinogenicity**

|                |            |
|----------------|------------|
| Carbon black : | Category 2 |
|----------------|------------|

**Reproductive toxicity**

|               |                |
|---------------|----------------|
| Epoxy resin : | Not classified |
|---------------|----------------|

**Specific target organ toxicity - Single exposure**

|                 |                                 |
|-----------------|---------------------------------|
| Calcium oxide : | Category 1 (respiratory organs) |
|-----------------|---------------------------------|

**Specific target organ toxicity - Repeated exposure**

|                 |                                 |
|-----------------|---------------------------------|
| Calcium oxide : | Category 1 (respiratory organs) |
| Carbon black :  | Category 1 (respiratory system) |

**12.ECOLOGICAL INFORMATION****Hazardous to the aquatic environment (Acute)**

|                |              |                           |              |                                                                           |
|----------------|--------------|---------------------------|--------------|---------------------------------------------------------------------------|
| Epoxy resin :  | EC50 (48H) : | Crustacea (Daphnia magna) | 1.7 mg/L     | (CERI/NITE Hazard Assessment (preliminary version) (2006) and SIDS, 2007) |
| Carbon black : | EC50 (72H) : | Algae (Scenedesmus)       | >10,000 mg/L | (SIDS, 2007)                                                              |
|                | EC50 (24H) : | Crustacea (Daphnia magna) | >5,600 mg/L  | (SIDS, 2007)                                                              |
|                | LC50 (96H) : | Fish (Dace)               | >1,000 mg/L  | (SIDS, 2007)                                                              |

**13.DISPORSAL CONSIDERATIONS****Waste from Residues**

Follow the all regulations in your country.

**Contaminated Packaging**

Follow the all regulations in your country.

**14.TRANSPORT INFORMATION**

|                                                                 |      |
|-----------------------------------------------------------------|------|
| UN Class                                                        | 9    |
| UN No                                                           | 3077 |
| Packing group                                                   | III  |
| <b>Specific Precautionary Transport Measures and Conditions</b> |      |
| Follow the all regulations in your country.                     |      |

**15.REGULATORY INFORMATION**

Follow the all regulations in your country.

**16.OTHER INFORMATION**

The information herein is given in good faith, but no warranty, express or implied, is made.